Practice 7 — Partitions & Inclusion-Exclusion

Solutions to Selected Problems

Problems 1, 7, 10, 12

Discrete Mathematics — QYSU

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1 Problem 1 — Partitions of a 5-Element Set into 3 Classes

Theory Recap: Stirling Numbers of the Second Kind

The **Stirling number of the second kind** $S(n,k)$ counts the number of ways to partition an n-element set into exactly k non-empty, pairwise disjoint subsets (called *classes* or *blocks*).

Recurrence relation:

$$S(n,k) = S(n-1,\, k-1) \;+\; k \cdot S(n-1,\, k), \qquad n \geq 1,\; 1 \leq k \leq n.$$

Interpretation: Consider element n.

- *n is alone in its own class:* the remaining $n-1$ elements form $k-1$ classes $\longrightarrow$ $S(n-1,\, k-1)$ ways.
- *n joins one of the existing k classes:* first partition the other $n-1$ elements into k classes ($S(n-1,\, k)$ ways), then choose which class n joins (k choices) $\longrightarrow$ $k \cdot S(n-1,\, k)$ ways.

Boundary conditions: $S(n,1) = 1,\;\; S(n,n) = 1,\;\; S(n,0) = 0$ for $n \geq 1$.

$n \backslash k$	1	2	3	4	5
1	1				
2	1	1			
3	1	3	1		
4	1	7	6	1	
5	1	15	25 $\longleftarrow S(5,3) = 25$	10	1

Problem Statement

In how many ways can a 5-element set be partitioned into 3 non-empty classes?

Solution

We need the Stirling number $S(5,3)$. Using the recurrence:

$$S(5,3) = S(4,2) + 3 \cdot S(4,3).$$

Computing $S(4,2)$:

$$S(4,2) = S(3,1) + 2 \cdot S(3,2) = 1 + 2 \cdot 3 = 7.$$

Computing $S(4,3)$:

$$S(4,3) = S(3,2) + 3 \cdot S(3,3) = 3 + 3 \cdot 1 = 6.$$

Final result:

$$S(5,3) = 7 + 3 \cdot 6 = 7 + 18 = 25.$$

Verification by explicit enumeration. Let $\{a, b, c, d, e\}$ be our set. A partition into 3 classes must have type $\{3, 1, 1\}$ or $\{2, 2, 1\}$ (the only integer partitions of 5 into 3 parts).

- **Type $\{3, 1, 1\}$:** Choose the 3-element block: $\binom{5}{3} = 10$ ways. The remaining 2 singletons form an unordered pair, giving 1 arrangement. Total: 10.
- **Type $\{2, 2, 1\}$:** Choose the singleton: $\binom{5}{1} = 5$ ways. Partition the remaining 4 elements into two pairs: $\frac{1}{2!}\binom{4}{2} = 3$ ways (divide by 2! since the two pairs are unordered). Total: $5 \times 3 = 15$.

Grand total: $10 + 15 = 25$. ✓

$$S(5,3) = \boxed{25}$$

2 Problem 7 — Divisibility with Inclusion-Exclusion

Theory Recap: Inclusion-Exclusion Principle

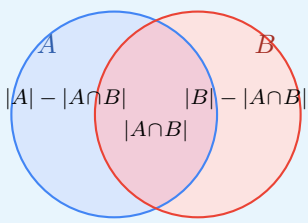
For events (or sets) $A_1, A_2, \ldots, A_m$ in a finite universe:

$$\left|\bigcup_{i=1}^{m} A_i\right| = \sum_i |A_i| - \sum_{i<j} |A_i \cap A_j| + \sum_{i<j<k} |A_i \cap A_j \cap A_k| - \cdots$$

To count elements that belong to **none** of $A_1, \ldots, A_m$:

$$\left|\overline{A_1} \cap \overline{A_2} \cap \cdots \cap \overline{A_m}\right| = |U| - \left|\bigcup_{i=1}^{m} A_i\right|.$$

Divisibility application: The number of integers in $\{1, 2, \ldots, N\}$ divisible by d is $\lfloor N/d \rfloor$. If $\gcd(d_1, d_2) = 1$, then divisibility by both d_1 and d_2 is the same as divisibility by $d_1 d_2$.



$$|A \cup B| = |A| + |B| - |A \cap B|$$

Problem Statement

How many integers between 1 and 16500 are not divisible by 5 and not divisible by 3, but **are** divisible by 11?

Solution

Let S be the set of integers in $\{1, 2, \ldots, 16500\}$ that are divisible by 11. We want to remove from S those elements also divisible by 3 or by 5.

Define:

$$\begin{aligned} A &= \{n \in S \mid 3 \mid n\} && \text{(divisible by both 11 and 3, i.e. by 33)}, \\ B &= \{n \in S \mid 5 \mid n\} && \text{(divisible by both 11 and 5, i.e. by 55)}. \end{aligned}$$

We seek $|S| - |A \cup B|$.

Step 1: Compute individual counts.

$$\begin{aligned} |S| &= \lfloor 16500/11 \rfloor &&= 1500, \\ |A| &= \lfloor 16500/33 \rfloor &&= 500, \\ |B| &= \lfloor 16500/55 \rfloor &&= 300, \\ |A \cap B| &= \lfloor 16500/165 \rfloor &&= 100. \end{aligned}$$

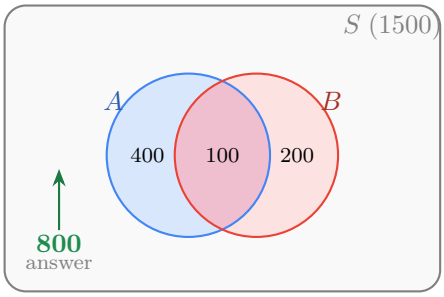
Here $|A \cap B|$ counts integers divisible by $\text{lcm}(33, 55) = 165$.

Step 2: Apply inclusion-exclusion.

$$|A \cup B| = |A| + |B| - |A \cap B| = 500 + 300 - 100 = 700.$$

Step 3: Subtract.

$$|S \setminus (A \cup B)| = |S| - |A \cup B| = 1500 - 700 = 800.$$



$\boxed{800}$ integers between 1 and 16500 are divisible by 11 but not by 3 and not by 5.

3 Problem 10 — Triangles in a Triangular Grid

Theory Recap: Counting Triangles in a Triangular Grid

When each side of an equilateral triangle is divided into n equal segments, connecting the division points with lines parallel to all three sides creates a **triangular grid** of side n.

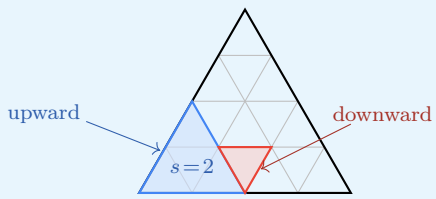
The grid contains two types of triangles with sides parallel to the original:

1. **Upward-pointing** triangles (same orientation as the original) of side length s ($1 \le s \le n$):

$$U(n,s) = \frac{(n-s+1)(n-s+2)}{2} = \binom{n-s+2}{2}.$$

2. **Downward-pointing** triangles (inverted orientation) of side length s ($1 \le s \le \lfloor n/2 \rfloor$):

$$D(n,s) = \frac{(n-2s+1)(n-2s+2)}{2} = \binom{n-2s+2}{2}.$$



Grid with $n = 4$

Problem Statement

Each side of triangle ABC is divided into 8 equal segments. Lines through the division points parallel to the sides create a triangular grid. How many triangles can be formed with vertices at these grid points such that their sides are parallel to the sides of ABC?

Solution

We have a triangular grid with $n = 8$. We count upward-pointing and downward-pointing triangles separately.

Upward-pointing triangles of side s ($1 \le s \le 8$):

$$U(8,s) = \frac{(8-s+1)(8-s+2)}{2} = \binom{10-s}{2}.$$

s	1	2	3	4	5	6	7	8	Total
$U(8,s)$	36	28	21	15	10	6	3	1	**120**

$$\sum_{s=1}^{8} U(8,s) = 36 + 28 + 21 + 15 + 10 + 6 + 3 + 1 = 120.$$

Downward-pointing triangles of side s $(1 \le s \le \lfloor 8/2 \rfloor = 4)$:

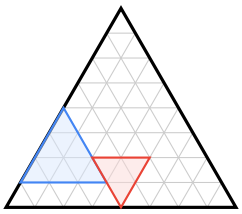
$$D(8,s) = \frac{(8-2s+1)(8-2s+2)}{2} = \binom{10-2s}{2}.$$

s	1	2	3	4	Total
$D(8,s)$	28	15	6	1	**50**

$$\sum_{s=1}^{4} D(8,s) = 28 + 15 + 6 + 1 = 50.$$

Grand total:

$$120 + 50 = 170.$$



$n = 8$ **grid**

The total number of triangles with sides parallel to ABC is $\boxed{170}$ (120 upward + 50 downward).

4 Problem 12 — Bounded Integer Solutions

Theory Recap: Counting Integer Solutions with Upper Bounds

To count non-negative integer solutions to $y_1 + y_2 + \cdots + y_r = n$ with upper bounds $y_i \le u_i$:

1. **Without upper bounds:** The number of solutions is $\binom{n+r-1}{r-1}$ (stars and bars).
2. **Subtract violations** using inclusion-exclusion. Let A_i be the set of solutions where $y_i > u_i$. Substituting $z_i = y_i - (u_i + 1) \ge 0$, the count $|A_i|$ becomes the unrestricted count with a reduced sum.
3. **Intersections:** $|A_i \cap A_j|$ substitutes both $z_i = y_i - (u_i + 1)$ and $z_j = y_j - (u_j + 1)$. If the remaining sum is negative, there are 0 solutions.

Formula:

$$\text{answer} = \sum_{S \subseteq \{1,\ldots,r\}} (-1)^{|S|} \binom{n - \sum_{i \in S}(u_i + 1) + r - 1}{r - 1},$$

where terms with a negative top entry in the binomial coefficient are 0.

Problem Statement

Find the number of integer solutions to

$$x_1 + x_2 + x_3 = 40, \qquad 4 \le x_1 \le 15, \quad 9 \le x_2 \le 18, \quad 5 \le x_3 \le 16.$$

Solution

Step 1: Shift variables to remove lower bounds.

Set $y_1 = x_1 - 4,\ y_2 = x_2 - 9,\ y_3 = x_3 - 5$. Then:

$$y_1 + y_2 + y_3 = 40 - 4 - 9 - 5 = 22,$$

with bounds $0 \le y_1 \le 11,\ 0 \le y_2 \le 9,\ 0 \le y_3 \le 11$.

Step 2: Count without upper bounds.

Without upper bounds, the number of non-negative integer solutions to $y_1 + y_2 + y_3 = 22$ is:

$$\binom{22+2}{2} = \binom{24}{2} = 276.$$

Step 3: Define violation sets and apply inclusion-exclusion.

Let A_i be the set of solutions where y_i exceeds its upper bound:

$$A_1 : \ y_1 \ge 12. \quad \text{Set } z_1 = y_1 - 12 \ge 0;\ z_1 + y_2 + y_3 = 10. \qquad |A_1| = \binom{12}{2} = 66.$$

$$A_2 : \ y_2 \ge 10. \quad \text{Set } z_2 = y_2 - 10 \ge 0;\ y_1 + z_2 + y_3 = 12. \qquad |A_2| = \binom{14}{2} = 91.$$

$$A_3 : \ y_3 \ge 12. \quad \text{Set } z_3 = y_3 - 12 \ge 0;\ y_1 + y_2 + z_3 = 10. \qquad |A_3| = \binom{12}{2} = 66.$$

Step 4: Pairwise intersections.

$A_1 \cap A_2$: $y_1 \geq 12,\ y_2 \geq 10$. Remaining sum: $22 - 12 - 10 = 0$. $\quad |A_1 \cap A_2| = \binom{2}{2} = 1$.

$A_1 \cap A_3$: $y_1 \geq 12,\ y_3 \geq 12$. Remaining sum: $22 - 12 - 12 = -2 < 0$. $\quad |A_1 \cap A_3| = 0$.

$A_2 \cap A_3$: $y_2 \geq 10,\ y_3 \geq 12$. Remaining sum: $22 - 10 - 12 = 0$. $\quad |A_2 \cap A_3| = \binom{2}{2} = 1$.

Step 5: Triple intersection.

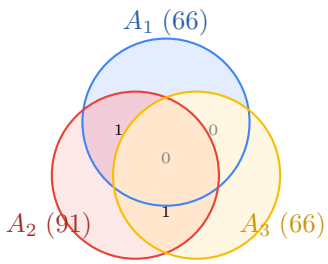
$$A_1 \cap A_2 \cap A_3 :\ y_1 \geq 12,\ y_2 \geq 10,\ y_3 \geq 12. \quad \text{Sum } \geq 34 > 22. \quad |A_1 \cap A_2 \cap A_3| = 0.$$

Step 6: Combine by inclusion-exclusion.

$$\begin{aligned} |A_1 \cup A_2 \cup A_3| &= (66 + 91 + 66) - (1 + 0 + 1) + 0 \\ &= 223 - 2 = 221. \end{aligned}$$

Final count:

$$276 - 221 = 55.$$



$|A_1 \cup A_2 \cup A_3| = 221$

Answer: $276 - 221 = 55$

The number of integer solutions is $\boxed{55}$.